

associates to *each* element x of X a *unique* element, say, y of Y . The unique element y associated with x of X is denoted as $f(x)$, i. e. $y=f(x)$.

Injective Function

- : A function $f: X \rightarrow Y$ is called injective, if for x_1, x_2 in X with $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. The definition can be equivalently stated as: A function $f: X \rightarrow Y$ is called injective, if $f(x_1) = f(x_2)$ then $x_1 = x_2$.

Partition of a set

- : For a given set X , a partition of X is a non-empty, finite or infinite, collection $\mathcal{P} = \{X_i\}$, where X_i 's are mutually disjoint non-empty subsets of X , with their union equal to whole X . More explicitly, a collection $\mathcal{P} = \{X_i\}$ is a partition of X , if
 - i) $X_i \neq \emptyset$, for each $X_i \in \mathcal{P}$,
 - ii) $X_i \cap X_j = \emptyset$, for every distinct pair of X_i and X_j in $\mathcal{P}$
 - iii) $\bigcup_{i \in \mathcal{P}} X_i = X$

Partial order relation

- : A relation R on a set X is said to be a **partial order**, if R is reflexive, *anti-symmetric* and transitive. In more detailed form, R is an equivalence relation on set X , if and only if
 - i) R is reflexive on X , i. e. for **every** $x \in X$, we have $(x, x) \in R$, and
 - ii) R is anti-symmetric on X , i. e. for $x, y \in X$, **if** $(x, y) \in R$, **and** $(y, x) \in R$, then $x=y$, and
 - iii) R is transitive on X , i. e. for $x, y, z \in X$, **if** $(x, y) \in R$, **and** $(y, z) \in R$, **then** $(x, z) \in R$.

Partially ordered set (Poset)

- : An ordered pair (X, R) —where X is non-empty set and R , a partial order relation on X —is called a partially ordered set (for short, poset). For example, $(N, \text{is-divisor -of})$ is a partially ordered set.

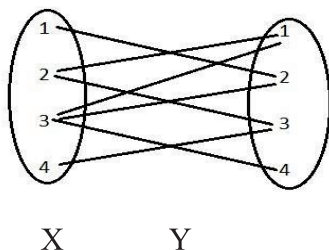
Surjective Function

- : A function $f: X \rightarrow Y$ is called surjective, if for an arbitrary element y of Y (*the codomain*), we can find an x in X (*the domain*) such that $f(x) = y$.

3.6 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1.i) Graphic Representation



ii) Matrix Representation

	1	2	3	4
1	0	1	0	0
2	1	0	1	0
3	1	1	0	1
4	0	0	1	0

2. i) The relation of *is-congruent-to* on the set X of all triangles is *symmetric*
- ii) The relation less-than-or-equal-to ($\leq$) on numbers is *not symmetric*, as $3 \leq 4$ is true, but $4 \leq 3$ is false. Also, $\leq$ is *anti-symmetric*, as $x \leq y$ and $y \leq x$, then $x = y$.
- iii) The relation *is-brother-of*, if underlying set X = set of all male human beings, is *symmetric*, as every male is its own brother (by definition, x is-brother-of y , if x is male and x and y have same parents). Also, it is *not anti-symmetric*, as for $x \neq y$, it is possible that we simultaneously have (x is-brother-of y) and (y is-brother-of x)

Check Your Progress 2

1. i) $\{(1, 1), (2, 2), (3, 3)\}$ is both *symmetric* and *antisymmetric*.
- ii) $\{(1, 2), (2, 3), (1, 3)\}$ is *antisymmetric*, but *not symmetric*.
- iii) $\{(1, 2), (2, 1), (1, 3), (3, 1)\}$ is *symmetric*, but *not anti-symmetric*, as $(1, 2), (2, 1)$, but $1 \neq 2$.

Check Your Progress 3

- 1) For each of the following relations, tell whether it is transitive, not-transitive, anti-transitive
- i) the relation R defined as R (a, b), for straight lines a and b, if and only angle between a and b is 30 degrees, is *NOT transitive*, because for R (a,

b) and $R(b, c)$, the angle between (lines a and b) and also (lines b and c) may be 30 degrees, but angle between a and c may be 60 degree also.

- ii) the relation less-than-or-equal-to ($\leq$) on numbers *is transitive*
- iii) The relation *is-brother-of*, if underlying set X = set of all human beings *is transitive*.

Check Your Progress 4

- 1) The three properties of relations on a set X , are **independent** of each other.
Examples of relation which is

- i) ***reflexive, but neither symmetric, nor transitive:***

R on Z , the set of integers, $(a, b) \in R$, if and only if $0 \leq a - b \leq 1$

Reflexive: $0 = a - a \leq 1$;

not symmetric: $(4, 3) \in R$, $(3, 4) \notin R$; not transitive: $(4, 3) \in R$, $(3, 2) \in R$, $(4, 2) \notin R$

- ii) ***symmetric, but neither reflexive, nor transitive:***

R , the relation of being perpendicular to each other in the set of all straight lines in a plane

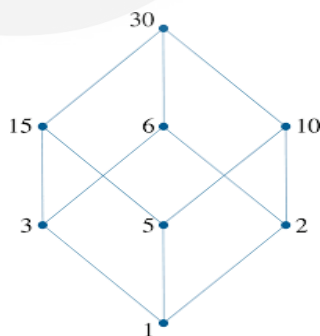
- iii) ***transitive, but neither reflexive, nor symmetric***

R is less-than ($<$) on the set of numbers, say, rational numbers

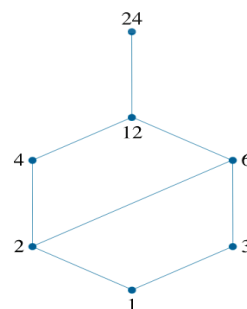
- 2) Partition of $X = \{\{1, 3\}, \{2, 4\}\}$

Check Your Progress 5

1.i)

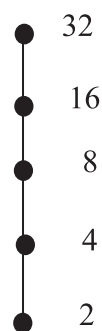
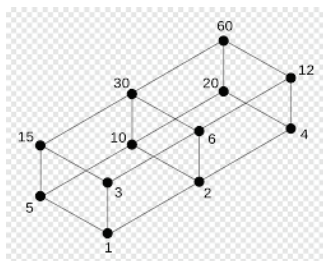


ii)



For (i) above, Minimal element = 1, Maximal element = 30;

For (ii) above, Minimal element = 1, Maximal element = 24



(iii)

(iv)

For (iii) above Minimal element = 1, Maximal element = 60;

For (iv) above Minimal element = 1, Maximal element = 32

Check Your Progress 6

1) i) Domain = $\mathbb{R}$, Range = $\{x \in \mathbb{R} \text{ and } x \leq 0\}$. (ii) Domain = $[0, 3]$, Range = $[0, 3]$.
Note: had the function been complex-valued complex function, i. e. $f: \mathbb{C} \rightarrow \mathbb{C}$, then Range = Domain = $\mathbb{C}$.

2) i) -5 , (ii) 9 (iii) -11

3) i) $f: \mathbb{N} \rightarrow \mathbb{N}$ is not a function, as $f(0) = -3 \notin \mathbb{N}$, the codomain of f .

ii) $f: \mathbb{N} \rightarrow \mathbb{Z}$, is a function.

3.7 EXERCISES

Q1. For each of the following relations, tell whether it is reflexive, not-reflexive or anti-reflexive

i) $R = \{(1, 1), (2, 2), (1, 3), (2, 3)\}$ on $X = \{1, 2, 3\}$

ii) The relation *is-brother-of*, if underlying set X = set of all human beings

Ans. As $(3, 3) \notin R$, R is not reflexive. It is not anti-reflexive as $(2, 2) \in R$.

i) For the relation *is-brother-of*, if underlying set X = set of all human beings, then the relation is not reflexive, because **if x is a female**, then $(x, x) \notin \text{is-brother-of}$, and also **not anti-reflexive**, because if **x is a male**, then $(x, x) \in \text{is-brother-of}$.

Q2. For each of the following relations on the set $X = \{1, 2, 3\}$, tell whether it is symmetric, not-symmetric or anti-symmetric

i) $\{(1, 1), (2, 2), (3, 3)\}$

ii) $\{(1, 2), (2, 3), (1, 3)\}$

iii) $\{(1, 2), (2, 1), (1, 3), (3, 1)\}$

Ans. i) $\{(1, 1), (2, 2), (3, 3)\}$ is *both symmetric and antisymmetric*.

ii) $\{(1, 2), (2, 3), (1, 3)\}$ is *antisymmetric, but not symmetric*.

iii) $\{(1, 2), (2, 1), (1, 3), (3, 1)\}$ is *symmetric, but not anti-symmetric*, as $(1, 2), (2, 1)$, but $1 \neq 2$.

Q3. For each of the following relations on the set $X = \{1, 2, 3\}$, tell whether it is transitive, not-transitive, anti-transitive

i) $\{(1, 1), (2, 2), (3, 3)\}$

ii) $\{(1, 2), (2, 3), (1, 3)\}$

iii) $\{(1, 2), (2, 1), (1, 3), (3, 1)\}$

Ans.

i) $\{(1, 1), (2, 2), (3, 3)\}$ is transitive

ii) $\{(1, 2), (2, 3), (1, 3)\}$ is transitive

iii) $\{(1, 2), (2, 1), (1, 3), (3, 1)\}$ is **not transitive**, as $(1, 2) \in R, (2, 1) \in R$, but $(1, 1) \notin R$.

Q4. Justify each by an example (i) A relation may be reflexive and symmetric, but not transitive (ii) A relation may be reflexive and transitive, but not symmetric (iii) A relation may be symmetric and transitive, but not reflexive.

Ans. Let R be a relation on $\{1, 2, 3\}$. Then

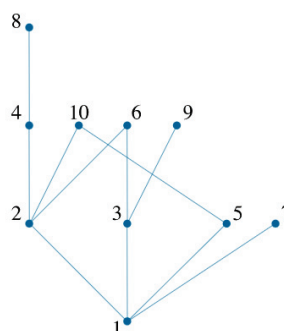
i) $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1)\}$ is a relation which is reflexive and symmetric, but not transitive, as $(2, 1)$ and $(1, 3) \in R$, but $(2, 3) \notin R$

ii) $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (2, 3)\}$ is a relation which is reflexive and transitive, but not symmetric, as $(2, 3) \in R$, but $(3, 2) \notin R$

iii) $R = \{(1, 1), (2, 2)\}$ is a relation which is symmetric and transitive, but not reflexive.

Q5. Under the relation of *is-proper-divisor-of*, draw Hasse diagram for set $X = \{1, 2, 3, \dots, 10\}$, and tell all the minimal elements, and maximal elements.

Ans. Minimal element = 1, Maximal elements = 5, 6, 7, 8, 9, 10.



Q6. Find whether the function $f: \mathbb{Z} \rightarrow \mathbb{R}$, defined by the rule $f(x) = x^2$ is injective?

Ans. Let $f(x_1) = f(x_2)$, i. e. $(x_1)^2 = (x_2)^2$ implies $x_1 = \pm x_2$, for x_1 and x_2 in $\mathbb{Z}$.
e. g. $f(5) = f(-5)$, hence, f is NOT injective.

Without the previous process, we could have directly used $f(5) = f(-5)$, to say, f is NOT injective.

Q7. Find whether the function $f: \mathbb{C} \rightarrow \mathbb{R}$, defined by the rule $f(x) = x^2$ is surjective?

Ans. Let $x \in \mathbb{R}$ be arbitrary element of $\mathbb{R}$. then $\sqrt{x} \in \mathbb{C}$, such that $f(\sqrt{x}) = (\sqrt{x})^2 = x$. Hence, f is surjective.



UNIT 4 COORDINATE GEOMETRY AND REPRESENTATION OF FUNCTIONS

Structure

- 4.0 Objectives
- 4.1 Introduction
 - 4.1.1 Vertical Line Test of Function
 - 4.1.2 Relationship between Equation and Function
- 4.2 Coordinate Geometry
 - 4.2.1 Cartesian Coordinate System
 - 4.2.2 Translating Geometric Figures to Algebraic Equations
- 4.3 Graphing Linear Functions
 - 4.3.1 Absolute Value Function
 - 4.3.2 Step Function
- 4.4 Graphing Non-Linear Functions
 - 4.4.1 Even and Odd Functions
 - 4.4.2 Quadratic Functions
 - 4.4.2.1 Computation of Vertex
 - 4.4.3 Cubic Function
- 4.5 Graphing Asymptomatic Functions
 - 4.5.1 Graph of Asymptote
 - 4.5.2 Square Root Function
 - 4.5.3 Exponential Function
 - 4.5.4 Logarithmic Function
- 4.6 Graphing Piecewise Functions
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- 4.7 Graphing Hyperbola and Parabola
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- 4.8 Graphing Functions with Two Variables
 - 4.8.1 Level Curves
- 4.9 Let Us Sum Up
- 4.10 Key Words
- 4.11 Answers/Hints to Check Your Progress Exercises
- 4.12 Exercises

4.0 OBJECTIVES

After going through this unit, you should be able to

- apply vertical line test to identify a non-function;
- explain how coordinate geometry differs from the conventional/synthetic geometry;
- obtain algebraic equations corresponding to simple geometric figures;
- graphically represent various types of functions of a single variable; and
- handle the graphs of two variable functions at a preliminary level.

4.1 INTRODUCTION

From the previous unit, we know that a function is a relationship between its domain (X , in Fig. 4.1A) and codomain (Y , in Fig. 4.1A) such that every element of the domain is related to *exactly one* element of the codomain/ range. On this ground, Fig. 4.1A does not represent a function (or functional relationship) on each of the following two grounds:

- the element 5 of domain (X) is related to three elements, viz. 4, 5, 7 of the codomain
- each of the elements 3 and 4 of domain X is not related to any element of the codomain Y .

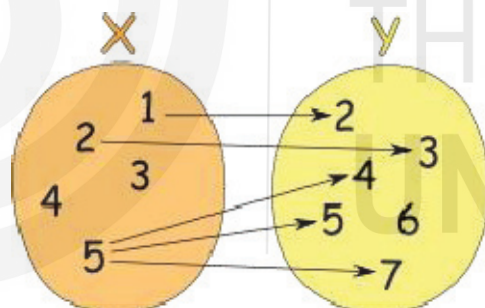


Fig. 4.1A: A Non-Functional Relationship

However, for a relationship to be a function, there is *no such restriction* on the elements of *codomain* Y . For example, Fig. 4.1B represents a function or functional relationship, even when

- element C of codomain Y corresponds to two elements viz. 2 and 3 of domain X , and
- element A (or even B) of codomain Y is not related to any element of domain X .

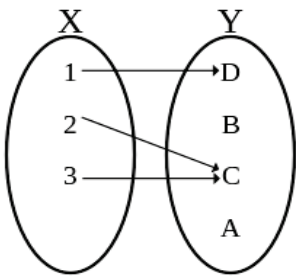


Fig. 4.1B: A Functional Relationship

4.1.1 Vertical Line Test of Function

The **vertical line test** for graphically testing *(only) failure* of a relationship $f(x)$ to be a function of x indicates that the vertical line crosses the graph of $f(x)$ at two points (Fig 4.2).

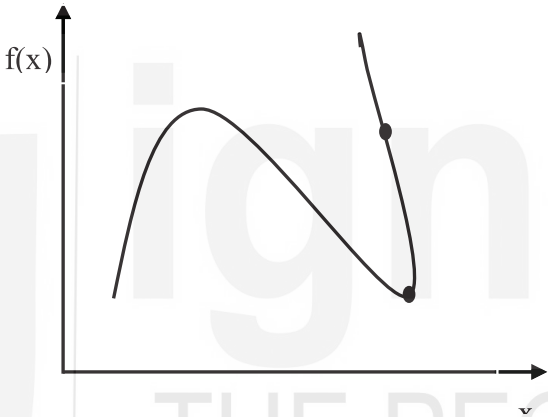
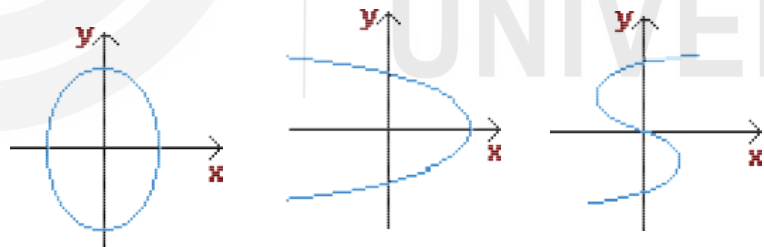


Fig. 4.2: Graph that does not Represent a Function

Thus, using vertical-line test we can identify the graphs that are not functions.



For example, you can show that each of the following graphs is not a function

Fig. 4.3: Graph of Not-Function

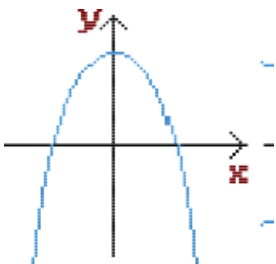


Fig. 4.4: Graph of a Function

But the graph shown in Fig 4.4 above represents a function. Thus, some of the important equations such as ellipse, circle and parabola may not represent a function.

One must note that the above test can be used *for testing failure* of a relationship to be functional. It cannot be used for testing (success) of a relationship to be functional. There are other additional requirements for a relationship to be functional.

4.1.2 Relationship between Equation and Function

Before we discuss functions in detail, we need to be clear about possible relationship between an equation and a relation/ function.

Consider the equation

$$X^2 + Y^2 = 4 \quad (1)$$

Then to talk about mathematical relation/function, we rewrite the above equation so that one of the variables, say, Y is written on L.H. S. and everything else is shifted to R.H.S.as follows:

$$Y^2 = 4 - X^2 \quad (2)$$

Next step to talk about a relation/ function corresponding to Equation (1) is to rewrite (2), so that only Y (and not its power, or square-root etc.) is written on L.H.S. as follows:

$$Y = \pm \sqrt{4 - X^2} \quad (3)$$

At this stage, after reducing the L.H.S. to a single variable, with the power of the variable as exactly 1, we can talk of a (mathematical) relation/function. As (3) does not satisfy single-value rule, hence, equation (1) does not correspond to a function. Of course, it corresponds to a relation. Equation (3) is obtained to make Y as dependent variable, and X as independent variable. By interchanging the roles of X and Y in the above process, we can make X as dependent variable, and Y as independent variable. Similarly, by taking the following equation in three variables

$$X^2 + Y^2 + Z^2 = 4 \quad (4)$$

Through the similar process, we may rewrite the Equation (4) as

$$Z = \pm \sqrt{4 - X^2 - Y^2} \quad (5)$$

At this stage, we may talk of Z being a function or not, and this time, function of two dependent variables, viz. X and Y. Again (5) does not represent a function.